MERN Syllabus

For Apprenticeship Training Program

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Skill India
कौशल भारत-कुशल भारत

N·S·D·C
National
Skill Development
Corporation
Transforming the skill landscape

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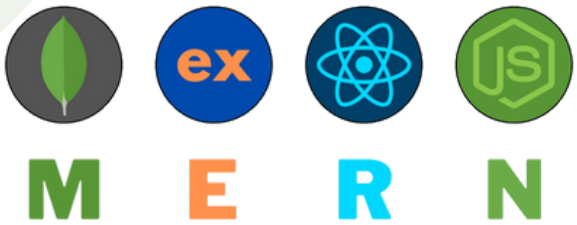
MERN stands for **MongoDB, Express.js, React.js, and Node.js.** These four technologies come together to form a powerful stack that allows developers to build dynamic, full-stack web applications with ease. Each component in the MERN stack serves a specific role, contributing to a seamless development process that is ideal for building both modern, user-friendly websites and robust, scalable web applications.

MongoDB is a NoSQL database that stores data in a flexible, JSON-like format. It allows developers to handle large amounts of data in a highly scalable and efficient way. Unlike traditional relational databases, MongoDB is schema-less, meaning that data can be easily modified and expanded as your application grows. This flexibility makes it a perfect choice for applications that require the storage and retrieval of unstructured or semi-structured data.

Express.js, a web application framework for Node.js, simplifies the process of building web applications by providing robust features for routing, handling HTTP requests, and managing middleware. Its minimalist approach allows developers to build scalable web applications and APIs quickly, without getting bogged down in the complexities of the server-side code. By using Express.js, developers can focus more on the logic and functionality of their applications rather than worrying about the infrastructure.

React.js, the front-end library in the MERN stack, is renowned for its ability to build fast, interactive, and highly dynamic user interfaces. Developed by Facebook, React allows developers to create reusable UI components that update efficiently as the application's data changes. Its virtual DOM ensures optimal performance, making React a preferred choice for building single-page applications (SPAs) and large-scale web projects that require dynamic, real-time user experiences.

Finally, **Node.js** is a powerful, event-driven JavaScript runtime that allows developers to run JavaScript on the server side. It enables fast and efficient execution of code, which is crucial for handling high volumes of requests in real-time applications. Node.js is built on Chrome's V8 engine, giving it exceptional performance capabilities. Moreover, its non-blocking I/O model makes it ideal for building scalable network applications, where thousands of connections can be handled simultaneously.



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HTML : MODULE 1

Introduction:

- What is HTML?
  1. Definition and importance of HTML
  2. Role of HTML in web development
  3. Overview of how HTML works with CSS and JavaScript
- History Of HTML ?
  1. Evolution of HTML from HTML 1.0 to HTML5
  2. Key milestones and developments in HTML history
  3. Impact of HTML advancements on web development
- Basic Structures of a Web Page
  1. Fundamental structure of an HTML document
  2. Understanding the <!DOCTYPE html> declaration
  3. Introduction to HTML elements and attributes

HTML Head Tags:

- Purpose and Use of the <head> Section
  1. Common head tags: <title>, <meta>, <link>, <style>, <script>, <meta viewport="">
  2. Using meta tags for SEO and responsive design

HTML Body Tags:

- Structure and Purpose of the <body> Section
  1. Creating and organising content within the body
  2. Basic HTML tags for structuring web pages

HTML Basic Tags:

-
 and <hr> Tags
  1. Usage and examples of line breaks with
  2. Creating horizontal rules with <hr>
- <p>, <h1> to <h6>, <a>, <img>, <ul>, <ol>, <li>, , , , , <table>, <tr>, <td>, <th>, <form>, <input>, <button>

HTML Link Creation:

- HTML Anchor Tags
  1. Creating hyperlinks with the <a> tag
  2. Understanding href attribute and link destinations
  3. Opening links in new tabs or windows with target attribute

HTML Tables

- HTML Tables
  1. Constructing tables with <table>, <tr>, <td>, and <th> tags
  2. Basic table structure and usage

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- HTML Nested Tables
  1. Creating tables within tables
  2. Managing nested table layouts
- HTML Merging Cells
  1. Merging table cells with colspan and rowspan attributes
  2. Practical examples of cell merging
- HTML Text Wrapping
  1. Wrapping text within table cells
  2. Handling overflow and text alignment
- HTML Table Background Image
  1. Adding background images to tables or cells
  2. Customising table aesthetics with background images
- HTML Table Cell Alignment
  1. Aligning text and content within table cells
  2. Using align and valign attributes

HTML : MODULE 2

HTML Lists:

- HTML Unordered Lists
  1. Creating bulleted lists with <ul>
  2. Using <li> for list items
  3. Styling unordered lists
- HTML Ordered Lists
  1. Creating numbered lists with <ol>
  2. List item customization and attributes
  3. Nested lists and their formatting

HTML Image Insertion:

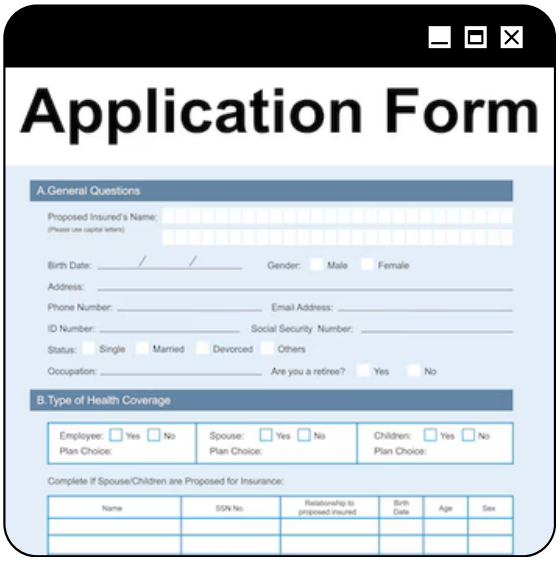
- Inserting Images with the <img> Tag
  1. Understanding src and alt attributes
  2. Image dimensions and alignment

HTML Embedding Videos:

- Embedding Videos Using the <video> Tag
  1. Video attributes: controls, autoplay, loop, muted
  2. Supporting different video formats

HTML Forms:

- Introduction to Forms
  1. Purpose and structure of HTML forms
  2. Basic form elements and their roles



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- HTML Form Tags and Attributes
  1. Essential form tags: <form>, <input>, <label>, <fieldset>
  2. Understanding form attributes: action, method, name, id
- HTML Forms - Post vs. Get
  1. Differences between POST and GET methods
  2. Appropriate use cases for each method
- HTML Forms - Input Text Fields
  1. Creating text input fields with <input type="text">
  2. Attributes for text fields: placeholder, maxlength, required
- HTML Forms - Select Menus
  1. Creating drop-down lists with <select> and <option> tags
  2. Handling multi-select menus
- HTML Forms - Checkboxes and Radio Buttons
  1. Using <input type="checkbox"> and <input type="radio">
  2. Grouping radio buttons for exclusive selection
- HTML Forms - Text Areas and Buttons
  1. Creating multi-line text input with <textarea>
  2. Adding buttons: <button>, <input type="submit">, <input type="reset">

TAILWIND CSS FUNDAMENTALS : MODULE 1

Introduction to Tailwind

- What is Tailwind CSS?
- Why use utility-first CSS?
- Comparison: Tailwind vs. traditional CSS

Setting Up Tailwind

- Create a basic HTML file
- Add Tailwind CDN link
- Verify Tailwind classes work (e.g., bg-red-500)

Working with Utility Classes

- Text & Font: text-lg, text-red-500, font-bold
- Background & Color: bg-blue-500, text-white, bg-gray-100
- Spacing: m-4, p-2, mt-8

Border and Radius Utilities

- Add borders: border, border-2
- Customize radius: rounded, rounded-full
- Border color & shade: border-gray-300, border-red-500

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TAILWIND CSS LAYOUT & POSITIONING : MODULE 2

Flexbox Layout

- Enable flex: flex
- Center content: justify-center, items-center
- Direction and gap: flex-col, gap-4

Grid Layout

- Basic grid: grid, grid-cols-2
- Responsive grids: md:grid-cols-3, lg:grid-cols-4
- Grid spacing: gap-2, gap-x-4, gap-y-6

Responsive Design

- Screen prefixes: sm:, md:, lg:
- Example use: text-sm md:text-lg
- Hide/show: hidden md:block

Position and Alignment

- Use relative, absolute
- Align content with text-center, text-right
- Center items vertically using flexbo

TAILWIND CSS INTERACTIONS & PROJECT : MODULE 3

Hover and Focus Effects

- Background hover: hover:bg-green-500
- Text hover: hover:text-blue-700
- Focus styles: focus:outline-none, focus:ring

Shadow & Transitions

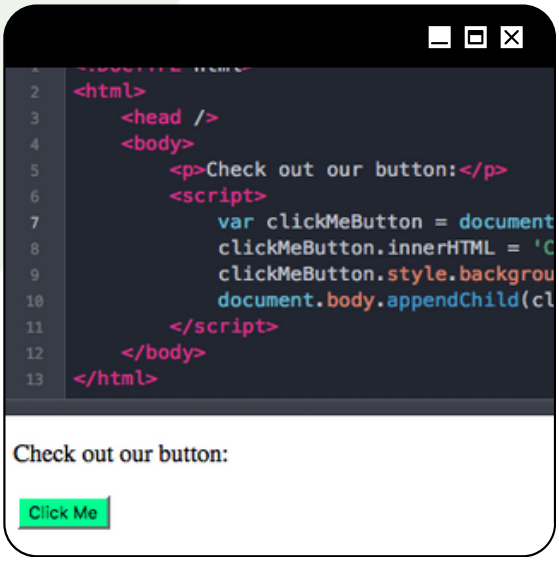
- Basic shadow: shadow, shadow-md
- On hover: hover:shadow-lg, hover:shadow-xl
- Smooth transition: transition, duration-300, ease-in-out

Button and Card Styling

- Buttons: bg-blue-500 text-white px-4 py-2
- Cards: p-6 shadow rounded-lg
- Hoverable cards: hover:bg-gray-100

Mini Project – Landing Page or Card

- Use layout utilities (flex/grid)
- Make it responsive with sm:, md:, lg:
- Add hover effects, spacing, and transitions



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JAVASCRIPT : MODULE 1

Introduction: In this module we will learn about what JavaScript is and the benefits of the language. We can also learn how to write the first JavaScript program.

- JavaScript Introduction
- First Javascript program
- Benefits of Javascript

JAVASCRIPT : MODULE 2

Language Syntax: In this module we will learn about JavaScript language syntax and how to use the program level. And we can also learn to understand the arrays.

- Variable declaration
- Operators
- Control Statements
- Error Handling
- Understanding arrays
- Function Declaration

JAVASCRIPT : MODULE 3

HTML Forms: In this module we will learn about how to use the forms. What are the properties in JavaScript? HTML form validation can be done by JavaScript.

- HTML Document object Model
- Working with HTML form and its elements

JAVASCRIPT : MODULE 4

HTML DOM: In this module we will learn about how JavaScript can access and change all the elements of an HTML document.

- HTML Document object Model
- Working with HTML form and its elements

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INTRODUCTION : MODULE 1

Introduction
What is the "MERN STACK"?
MERN - A FIRST OVERVIEW
Course Outline
How To Get The Most Out Of The Course
Using the Course Resources
Course Setup

THE MERN STACK : MODULE 2

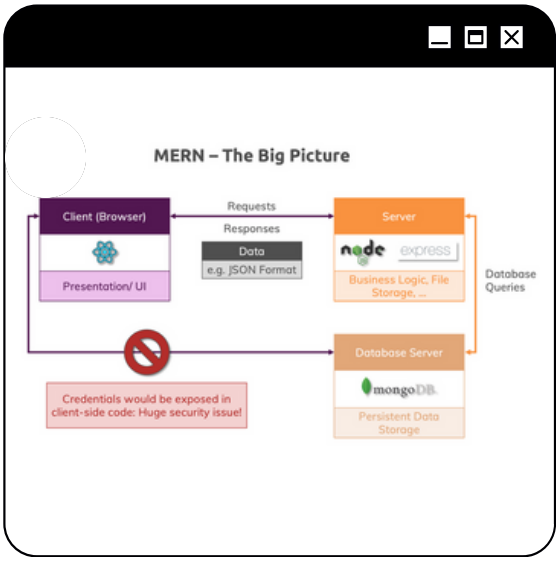
Module Introduction
Understanding the Big Picture
Diving Into the Frontend
Understanding the Backend
REST vs GraphQL
Connecting Node & React
MERN - Essentials
Creating our Development Environment & the Development Servers
Diving Deeper Into the Code
Useful Resources & Links

PLANNING THE APP : MODULE 3

Module Introduction
Understanding the General App Idea
Sketching out the Frontend
Data & API Endpoints used in our App
Required SPA Pages for the Frontend

REACT.JS - A REFRESHER : MODULE 4

Module Introduction
What is React?
A Note About The NodeJS Version
React 18
Setting Up a Starting Project
Understanding JSX
Understanding Components
Working with Multiple Components
Using Props to pass Data between Components
Rendering Lists of Data
Handling Events
Efficient Child<=>Parent Communication
Working with "State"
More on State
Fetching User Input (Two-way Binding)



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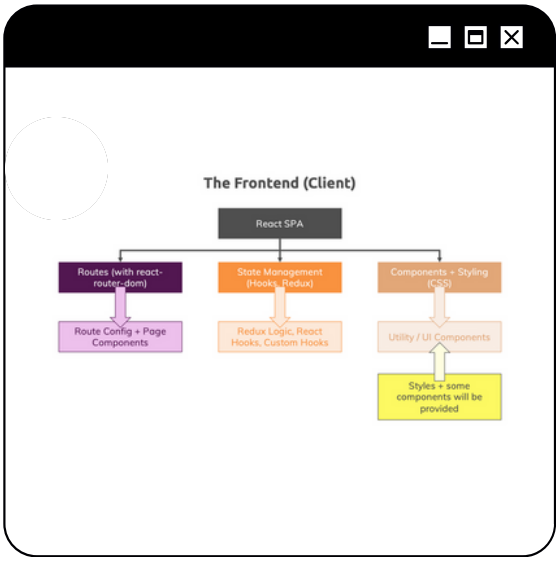
React.js Basics
Useful Resources & Links

REACT.JS - BUILDING THE FRONTEND : MODULE 5

Module Introduction
Important: Use React Script v5
Starting Setup, Pages & Routes
Adding a UsersList Page / Component
Adding a UserItem Component
Styling our App & More Components
Presentational vs Stateful Components
Adding a Main Header
Adding Navlinks
Implementing a Basic Desktop & Mobile Navigation
Understanding Portals
Handling the Drawer State
Animating the Sidedrawer
Rendering User Places & Using Dynamic Route Segments
Getting Route Params
Adding Custom Buttons
Adding a Modal
Rendering a Map with Open Streets Maps
Continuing without a Credit Card
Optional: More on the useEffect() Hook
Adding a Custom Form Input Component
Managing State in the Input Component
Adding Input Validation
Sharing Input Values & Adding Multiple Inputs
Managing Form-wide State
Finishing the "Add Place" Form
Starting Work on the "Update Place" Page
Adjusting the Input Component
Creating a Custom Form Hook (useForm)
Optional: More on (Custom) React Hooks
Adjusting the Form Hook
Fixing Minor Issues
Showing a Deletion Warning
Adding an "Auth" Page & Login Form
Adding Signup + "Switch Mode" Button
Adding Auth Context for App-wide State Management
Listening to Context Changes
Adding Authenticated & Unauthenticated Routes
More Auth Context Usage
Useful Resources & Links



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NODE.JS & EXPRESS.JS - A REFRESHER : MODULE 6

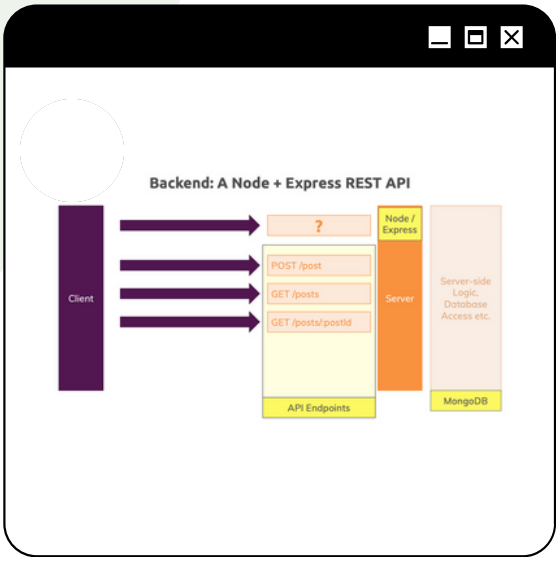
Module Introduction
What is Node.js?
Writing our First Node.js Code
Sending Requests & Responses
What is Express.js?
Adding Express.js
Understanding the Advantages of Express.js
How Code Execution Works
Node & Express Basics
Useful Resources & Links

NODE.JS & EXPRESS.JS - BUILDING OUR REST API : MODULE 7

Module Introduction
Setting up our Project
Implementing Basic Routing
Adding Place-Specific Routes
Getting a Place By User ID
Handling Errors
Adding our own Error Model
Adding Controllers
Adding a POST Route & Using Postman
Handling Errors for Unsupported Routes
Adding Patch Routes to Update Places
Deleting Places
Finalizing the "Get Places by User ID" Resource
Setting Up the User Routes (Signup, Login, Get Users)
Validating API Input (Request Bodies)
Validating Patch Requests & User Routes
Useful Resources & Links

WORKING DEEP WITH MONGO DB & MONGOOSE : MODULE 8

Module Introduction
What is MongoDB?
SQL vs NoSQL
Connecting React to a Database?
Setting Up MongoDB
Creating a Simple Backend & with the Database
Creating a Document with MongoDB
Getting Data from the Database
Installing Mongoose
Understanding Models & Schemas
Creating a Product



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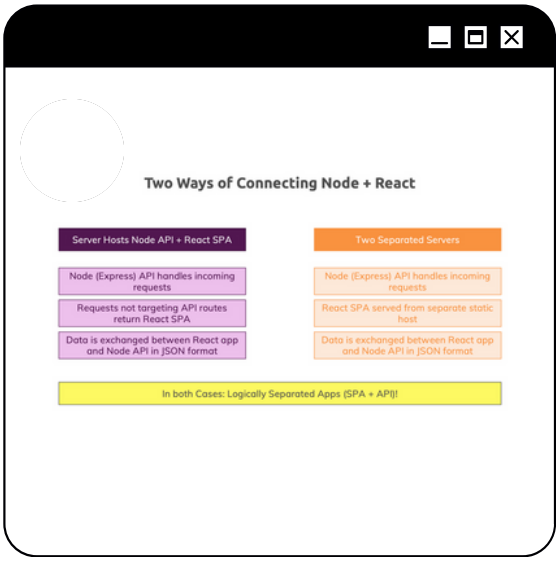
Connecting to the Database & Saving the Product
Getting Products
Understanding the ObjectID

CONNECTING THE BACKEND TO THE DATABASE - MONGO DB & MONGOOSE : MODULE 9

Module Introduction
Installing Mongoose & Connecting our Backend to MongoDB
Creating the Place Schema & Model
Creating & Storing Documents in the Database
Getting Places by the PlaceID
Getting Places by the UserID
Updating Places
Deleting Places
How Users & Places are Related
Creating the User Model
Using the User Model for Signup
Adding the User Login
Getting Users
Adding the Relation between Places & Users
Creating Places & Adding it to a User
Deleting Places & Removing it From the User
Getting Places - An Alternative
Cleaning Up our Code
Useful Resources & Links

CONNECTING THE REACT.JS FRONTEND TO THE BACKEND : MODULE 10

Module Introduction
Initial Setup
Sending a POST Request to the Backend
Optional: The fetch() API
Handling CORS Errors
Getting Started with Error Handling
Proper Error Handling in the Frontend
Sending a Login Request
Getting Users with a GET Request
Creating a Custom Http Hook
Improving the Custom Http Hook
Using the Http Hook to GET Users
Adding Places (POST)
Loading Places by User Id
Updating Places
Deleting Places
Fixing NavLinks & "My Places"
Final Adjustments
Useful Resources & Links



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ADDING FILE UPLOAD : MODULE 11

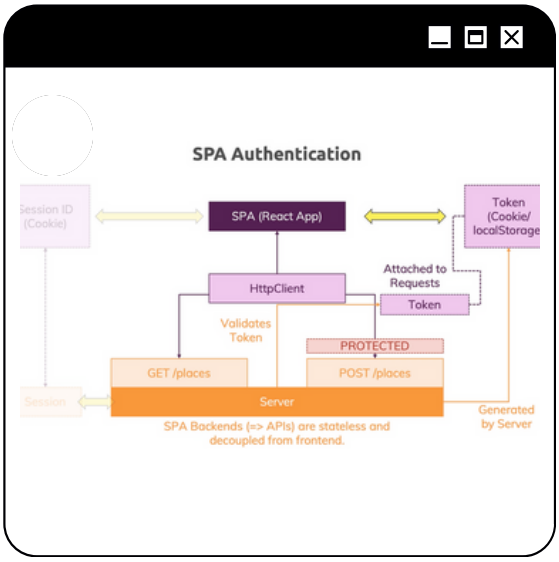
Module Introduction
Building an ImageUpload Component
Finishing & Using the ImageUpload Component (in a Form)
On the Backend: Using Multer to Save Files
Filtering Files on the Backend (Images Only!)
Wiring Frontend and Backend Up
Connecting Users to Images
Serving Images Statically
Uploading Images for New Places
Deleting Images When Places Get Deleted
Useful Resources & Links

ADDING AUTHENTICATION : MODULE 12

Module Introduction
How Authentication Works (in a MERN App)
Hashing the User Password
Logging Users In (with Hashed Passwords)
Generating Tokens (JWT) on the Backend
Optional: More on JSON Web Tokens
Backend Route Protection with Auth Middleware
Using & Attaching JWT (Tokens) in React
Using Tokens to Update and Delete Places
Adding Authorization
Authorization on "Delete Places"
Storing the Token in the Browser Storage
Adding Auto-Login (Basic Version)
Managing the Token Expiration Date
Finished Auto-Login & Auto-Logout
Creating a Custom Authentication Hook
Authentication - Summary
General App Improvements
Authentication & Authorization

APPLICATION DEPLOYMENT : MODULE 13

Module Introduction
Deployment Steps Overview
Using Environment Variables (Node.js)
Environment Variables in React Apps
Preparing API Keys & Credentials
Building the React App
Adding Code Splitting to React
Understanding Possible Deployment Alternatives
Deploying a Standalone REST API (Example: Heroku)



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WHAT IS AN AI IMAGE GENERATOR?

An AI image generator uses machine learning models to create images based on textual descriptions or other input data.

OVERVIEW OF THE MERN STACK FOR THE PROJECT

The MERN stack consists of MongoDB (database), Express (backend framework), React (frontend framework), and Node.js (runtime environment).

SETTING UP THE FOLDER STRUCTURE FOR A MERN PROJECT

The folder structure includes separate directories for backend (server), frontend (client), and shared components like utilities or assets.

INSTALLING NECESSARY DEPENDENCIES FOR THE BACKEND AND FRONTEND

Use npm to install dependencies like Express for the backend, and React along with MUI for the frontend.

CREATING A GIT REPOSITORY FOR VERSION CONTROL

Initialize a Git repository and commit your project to maintain version control and collaborate efficiently.

BACKEND DEVELOPMENT (NODE.JS + EXPRESS + MONGODB)

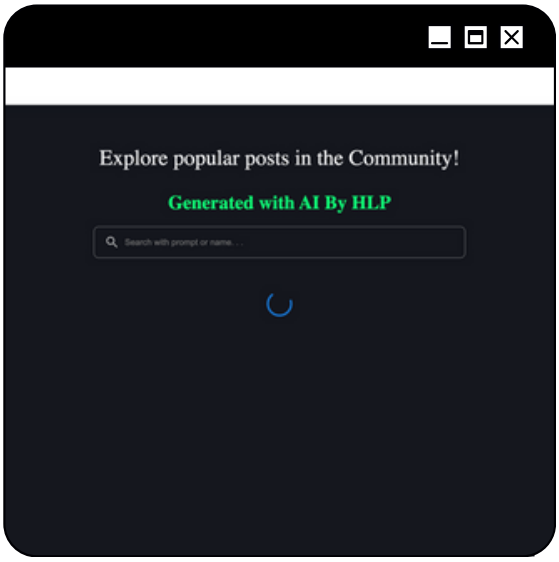
Set up Node.js and Express to create the backend API that communicates with MongoDB to store and retrieve data.

SETTING UP UNSPLASH API, CLOUDINARY

Obtain credentials for Unsplash API and Cloudinary, and securely store MongoDB credentials to access the services.

CREATING API ENDPOINTS FOR FETCHING AND STORING IMAGES

Implement API endpoints to interact with the image sources (Unsplash and Cloudinary), allowing image fetch and storage operations.



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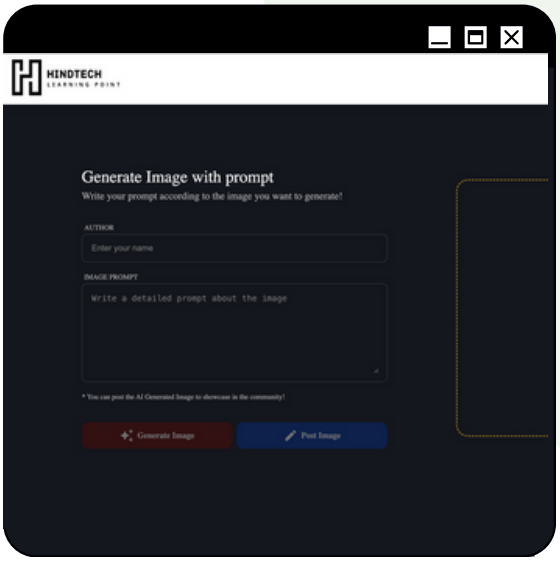
PROJECT DEVELOPMENT

1. **AI Chat & Gen (MAJOR PROJECT 1)**
2. **YOUR PLACES (MAJOR PROJECT 2)**

DURATION : 180 DAYS

OUTCOME:

- **Build Fullstack MERN Apps:** Master the MERN stack (MongoDB, Express, React, Node) to create powerful fullstack applications from scratch.
- **Add Authentication & File Uploads:** Learn how to implement user authentication, authorization, and file uploads for real-world web applications.
- **Deploy Fullstack Projects:** Understand various deployment strategies to take your MERN stack applications live on different hosting platforms.
- **Master Database Integration with MongoDB:** Understand how to effectively integrate MongoDB with Node.js and Express to manage data seamlessly in fullstack web applications
- **Real-Time Features with Socket.io:** Learn how to integrate real-time functionalities such as chat systems.



MAJOR PROJECT 1

MAJOR PROJECT 2

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MERN APPRENTICESHIP TRAINING SYLLABUS PROGRESS TRACKER

MERN

- ◯ Module 1
- ◯ Module 2
- ◯ Module 3
- ◯ Module 4
- ◯ Module 5
- ◯ Module 6
- ◯ Module 7
- ◯ Module 8
- ◯ Module 9
- ◯ Module 10
- ◯ Module 11
- ◯ Module 12
- ◯ Module 13

PROJECTS

- ◯ AI Chat & Gen
- ◯ YOUR PLACES

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NOTES

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CSE/IT or interested student
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Training	Duration	Amount
MERN Apprenticeship Training Program	6 to 8 Months	₹25,000

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We offer placement assistance to all eligible trainees.